

Multispectral Imagery for Wind Turbine (Windmill) Applications

Project Scope Document — with Open-Source Technology Stack and Sample Outputs

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1. Purpose & Objectives

This document defines the scope of a system that uses multispectral imagery (visible, red-edge, near-infrared and optionally SWIR/thermal bands) across the lifecycle of wind turbines: site selection, blade and structural inspection, environmental monitoring, and operations & maintenance (O&M) optimization. The solution is built entirely on open-source software.

- Automate detection of blade defects (erosion, cracks, moisture ingress, lightning damage) from UAV multispectral captures.
- Produce site-analysis products (NDVI, land cover, soil/water masks) from open satellite data (Sentinel-2, Landsat).
- Track vegetation and environmental change around wind farms for permit compliance.
- Deliver georeferenced defect maps, severity classifications and inspection reports through an open-source GIS/web stack.

2. In-Scope Work Packages

WP	Work Package	Description
WP1	Data acquisition	UAV multispectral flights (blade & tower); ingest of Sentinel-2/Landsat scenes for site & environment products.
WP2	Preprocessing	Radiometric calibration, band alignment/registration, orthorectification, mosaicking, reflectance conversion.
WP3	Index & analytics engine	NDVI/NDRE/EVI computation, anomaly masks, change detection between inspection epochs.
WP4	Defect detection (ML)	CNN / classical CV models for crack, erosion, moisture and lightning-damage classification with severity grading.
WP5	GIS & reporting	Georeferenced defect layers, web map dashboard, automated PDF inspection reports.
WP6	O&M integration	Export to CMMS/digital-twin via open APIs; trend analysis across inspection cycles.

Out of scope (phase 1): SCADA data fusion, offshore marine-growth monitoring, real-time onboard (edge) inference, LiDAR integration.

3. Open-Source Software Stack

Layer	Tool (License)	Role in this project
Satellite data access	ESA Sentinel-2 / USGS Landsat (free & open data); sentinelsat, pystac-client (GPL/Apache)	Free multispectral scenes and API download for site & environmental products.
Photogrammetry / stitching	OpenDroneMap / WebODM (AGPL-3.0)	Orthomosaics and 3D reconstruction from UAV multispectral flights, incl. band alignment for MicaSense-class sensors.

Layer	Tool (License)	Role in this project
Raster / geospatial core	GDAL (MIT), Rasterio (BSD), rioxarray (Apache-2.0)	Reading, reprojecting, calibrating and writing multispectral rasters.
Scientific computing	NumPy, SciPy, xarray (BSD)	Band math: NDVI, NDRE, EVI, custom anomaly indices.
Image processing / CV	OpenCV (Apache-2.0), scikit-image (BSD)	Denoising, thresholding, morphology, contour extraction for defect masks.
Machine learning	PyTorch (BSD) or TensorFlow (Apache-2.0); scikit-learn (BSD); Ultralytics YOLO (AGPL-3.0)	Defect detection/classification models; random-forest land-cover classification.
Semantic segmentation	segmentation-models-pytorch (MIT)	U-Net/DeepLab blade-surface defect segmentation.
GIS desktop	QGIS (GPL-2.0) + Semi-Automatic Classification Plugin	Visual QA, manual labelling, land-cover classification workflows.
Spatial database	PostGIS on PostgreSQL (GPL/PostgreSQL licenses)	Storage of defect geometries, turbine assets, inspection epochs.
Web mapping / dashboard	GeoServer (GPL-2.0), Leaflet (BSD), Streamlit (Apache-2.0)	Serving tiles and interactive inspection dashboards.
Visualization / reports	Matplotlib (PSF-based), ReportLab (BSD)	Figures and automated PDF inspection reports (this document was produced with them).
Workflow orchestration	Apache Airflow (Apache-2.0)	Scheduled ingestion, processing and report pipelines.

4. Sample Outputs (Synthetic Demonstration Images)

The images below are synthetic samples generated with NumPy + Matplotlib to illustrate the expected products. Production outputs will use real UAV and Sentinel-2 data processed through the stack in Section 3.

4.1 Site analysis product — multispectral bands and NDVI

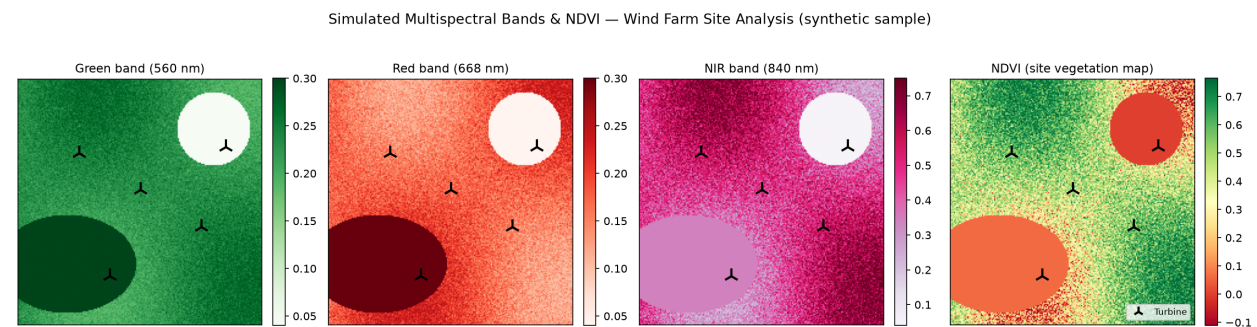


Figure 1 — Simulated green/red/NIR reflectance and derived NDVI over a wind farm site. Turbine symbols mark asset locations; low-NDVI areas (water body, bare soil) are automatically excluded from vegetation-compliance monitoring.

4.2 Blade inspection product — NIR capture and automated defect mask

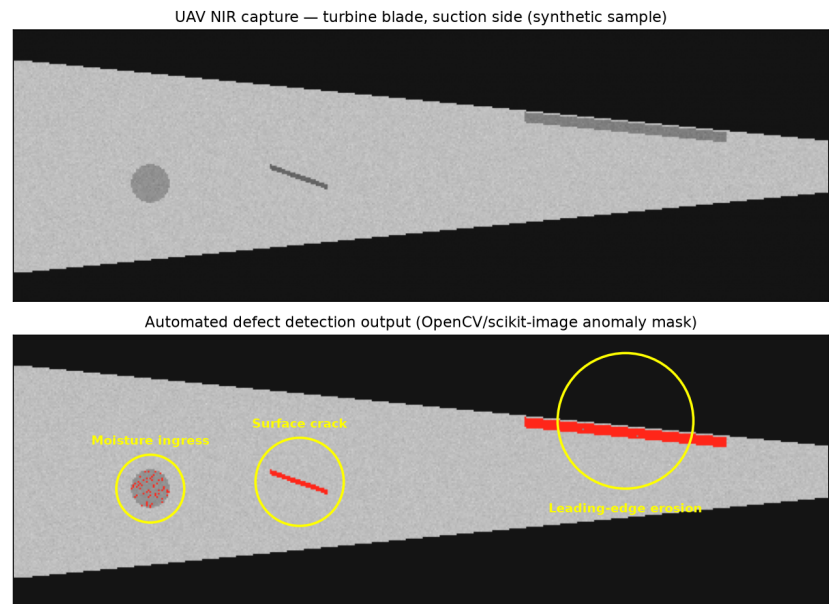


Figure 2 — Simulated UAV NIR image of a blade (top) and the automated anomaly detection overlay (bottom): leading-edge erosion, a surface crack and a moisture-ingress zone are flagged in red with yellow region-of-interest markers.

5. Processing Pipeline

Step	Stage	Open-source tools
1	Acquire — UAV flight plans / satellite scene download	OpenDroneMap ground control, sentinelsat/pystac-client
2	Calibrate — radiometric correction, reflectance panels	OpenDroneMap, Rasterio, NumPy
3	Register & mosaic — band alignment, orthomosaic	OpenDroneMap/WebODM, GDAL
4	Analyze — indices (NDVI/NDRE), anomaly masks	NumPy, xarray, OpenCV, scikit-image
5	Detect — ML defect classification & severity grading	PyTorch, segmentation-models-pytorch, YOLO
6	Store & serve — geodata layers, tiles, dashboard	PostGIS, GeoServer, Leaflet, Streamlit

Step	Stage	Open-source tools
7	Report — automated inspection PDFs, trend charts	Matplotlib, ReportLab, Airflow (scheduling)

6. Deliverables

- D1 — Calibrated multispectral orthomosaics per turbine/site (GeoTIFF).
- D2 — Defect detection layer with severity classes (GeoPackage/PostGIS).
- D3 — NDVI / land-cover / change-detection maps for environmental compliance.
- D4 — Web dashboard (Leaflet/Streamlit) with per-turbine inspection history.
- D5 — Automated PDF inspection reports (per flight campaign).
- D6 — Reproducible open-source pipeline (containerized, Apache Airflow DAGs).

7. Standards, Assumptions & Constraints

- Inspection practice aligned with IEC 61400 series and DNV-GL blade inspection guidance.
- UAV operations subject to local aviation (e.g. DGCA/FAA/EASA) rules; flights only in permitted wind conditions.
- Multispectral sensor with at least 5 bands (B, G, R, RedEdge, NIR); SWIR/thermal optional add-ons.
- All software components are OSI-approved open source; AGPL components (WebODM, YOLO) used as services to keep licensing obligations contained.
- Sample images in this document are synthetic and for illustration only.